

Exact tree-level Majorana kernels

A correction to the dimension-seven loop interpretation

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What this release establishes

The finite-pole tree kernel has an exact local expansion, an explicit remainder and a reconstruction from a finite set of moments when the heavy-mass nodes are known. These mathematical identities survive the review.

What this release corrects

The earlier local N00AK package treated a dimension-seven mixed-bubble cancellation as a derived physics result. Its argument multiplied a dimension-five loop result by one external momentum-squared factor. That step does not account for the different momentum routes of the derivative vertex. The dimension-seven coefficient and its higher-order extensions remain unresolved.

The cancellation tests verify the proposed matrices **if those matrices are assumed**. They do not derive the loop coefficient. This distinction is applied throughout the revised technical note and numerical package.

Current status

Research note and reproducibility release. No completed Theory of Everything, new experimental validation, or independently established dimension-seven loop coefficient is claimed.

The preceding published record is DOI 10.5281/zenodo.20241403, the May 16 epsilon-lattice mathematical audit; its title and version were verified on Zenodo on 4 September 2026. This later Type-I EFT branch does not establish a physical bridge from that construction to a complete ultraviolet theory.

Full derivations, conventions, primary references and the off-shell calculation plan are in TOE_N00AK_r1_TECHNICAL_NOTE.md.

1. Exact tree result

Use positive heavy masses M_d and complex Yukawa columns y_d . The superscript T means transpose, not Hermitian conjugate. Retain the lepton-number-violating mass-numerator part of tree exchange:

$$K_D(z) = \sum_d (y_d y_d^T / M_d) / (1 - z/M_d^2)$$

$$W_n = \sum_d y_d y_d^T / M_d^{2n+1}$$

Here z has mass dimension two. K has mass dimension minus one. For $|z|$ below the smallest removed mass squared, the geometric expansion converges:

$$K_D(z) = \sum_{n \geq 0} z^n W_n$$

After keeping terms through $n = N$, the exact remainder is

$$R_N(z) = \sum_d (y_d y_d^T / M_d) (z/M_d^2)^{N+1} / (1 - z/M_d^2).$$

The finite-remainder identity is algebraically valid away from the poles. The infinite series needs the convergence condition above. An absolute spectral-norm bound follows from submultiplicativity; the technical note states it explicitly.

Reconstruction with known nodes

Combine equal-mass residues and remove only nodes with exactly zero combined residue. If c effective nodes are known, define

$$P(z) = \prod_{a=1 \dots c} (1 - z/M_a^2) = \sum_{k=0 \dots c} p_k z^k, p_0 = 1.$$

Then $P(z)K_D(z)$ is a matrix polynomial of degree at most $c - 1$. The moments obey

$$\sum_{k=0 \dots c} p_k W_{n-k} = 0, n \geq c.$$

The first c moments reconstruct the rational kernel. This applies classical rational generating-function algebra. Unknown nodes require additional information, and nearly coincident nodes can make inverse recovery poorly conditioned. None of these identities determines a loop mixing coefficient.

Physical setting: Elgaard-Clausen and Trott, "On expansions in neutrino effective field theory," Sections 3–4, [arXiv:1703.04415](https://arxiv.org/abs/1703.04415).

2. Benchmark and test scope

The frozen benchmark uses masses $(2, 4, 8) \times 10^{12}$ GeV and the supplied full-precision complex Yukawa matrix. Errors below are relative Frobenius errors in the selected tree kernel at $z = M_{\text{probe}}^2$.

Removed modes	Probe	Dimension 5	Dimensions 5 + 7
N3	N2	25%	6.25%
N3	N1	6.25%	0.390625%
N2	N1	25%	6.25%
N3 + N2	N1	22.1643102559%	5.62145822425%

For a single nonzero pole and $r = M_{\text{probe}}/M_D < 1$, the exact relative errors are r^2 and r^4 , respectively. The two-pole row depends on the frozen complex matrix. These numbers are not uncertainties on a measured neutrino mass or on a completed one-loop prediction.

Checks and their limits

Tree checks. The public numerical runner exercises geometric remainders, finite-pole reconstruction, moment recurrence, the norm bound and the frozen benchmark.

Conditional algebra checks. A separate group varies the proposed survivor Yukawa map and compares it with the assumed contact. The tests use complex matrices and an arbitrary common coefficient, including values other than 3.

Negative control. Changing only the contact coefficient breaks the assumed relation. This checks that the numerical test detects inconsistency; it does not select the physical value of a common coefficient.

Provenance. The release records input and payload hashes plus the runtime used for fresh results. A checksum proves file identity. A numerical PASS proves only the named mathematical tests.

The original scripts passed fresh threshold checks in the audit. Rebuilding the unchanged original payload reproduced both source ZIPs exactly. Regenerating numerical output on a newer runtime changed some last digits and derived hashes; this is not described as a byte-identical historical scientific replay.

3. Conditional cancellation

Let Y and M describe surviving modes, with M positive and diagonal. At fixed M define

$$S = YM^{-1}Y^T, P_2 = YM^2Y^\dagger, A_2 = P_2W_1 + W_1P_2^T.$$

For any shared scalar c , assume the paired variations

$$\delta Y = -c W_1 Y^* M^3, \delta C_5 = c A_2.$$

Direct variation of S then gives

$$\delta S = \delta Y M^{-1}Y^T + YM^{-1}\delta Y^T = -c A_2,$$

$$\delta(C_5 + S) = 0.$$

The implication is exact for $c = 3$ and for other real or complex c . Cancellation cannot identify the physical loop coefficient because it holds for every common coefficient in the assumed pair. The corresponding higher-odd and rationally summed formulas have the same conditional status.

Where the derivation needs repair

The dimension-five reference has direct and crossed $SU(2)$ pairings. For the derivative operator, the total momentum of the differentiated lepton–Higgs composite differs between those pairings. One pair is entirely external; the crossed pair includes a loop momentum.

Therefore the full dimension-five answer cannot simply be multiplied by one common external p^2 without evaluating those momentum-dependent contributions. This is a diagnosis of an unsupported derivation step. It neither computes a replacement coefficient nor proves that a completed calculation cannot cancel.

The known dimension-five calculation remains separate. Applying sourced equations of motion consistently is necessary, but the off-shell pole must first be derived with all relevant momentum structures.

Primary reference: Di Zhang, “Threshold Effects on the Massless Neutrino in the Canonical Seesaw Mechanism,” Appendix A, [arXiv:2405.18017v3](https://arxiv.org/abs/2405.18017v3). The derivative-pairing diagnosis is an inference in this release, not a published dimension-seven result from that paper.

4. Next calculation and release use

The immediate research task is the routed off-shell W_1 mixed bubble. It has not been completed in this release.

- 1. Lock conventions.** State the Lagrangian, Fourier signs, Majorana rules, regulator, subtraction scheme and Green-operator basis. Reproduce the published dimension-five normalization first.
- 2. Retain every pairing.** Use independent external lepton and Higgs momenta. Write the derivative momentum polynomial for every direct and crossed SU(2) contraction before integration.
- 3. Evaluate the pole off shell.** Reduce the loop tensor integrals and document how ultraviolet poles are distinguished from infrared singularities. Keep every independent local momentum structure.
- 4. Match and reduce consistently.** Match the Green-basis counterterms, then apply the complete sourced equations of motion, including contact, mass and field variations where needed.
- 5. Compare and reproduce.** Evaluate the resulting contact and survivor-map variation. Test the proposed coefficient only now. Reproduce the calculation with a separate implementation and archive conventions, integrands and reductions.

Beyond this bubble

A complete sequential prediction still requires the remaining dimension-seven operator classes, lepton-number-conserving and time-ordered contributions, hard and finite threshold matching, running, and comparison with observables. No cross-project equations or fitted numbers have been imported as a substitute for a field-theory bridge.

Files and responsibility

The public archive includes the technical note, focused code, the frozen benchmark, fresh result files, provenance and checksums. Nested parent archives, private chat exports and screenshots are omitted. The original local research files are preserved unchanged.

AI-assisted drafting, code adaptation, mathematical review and checks used OpenAI Codex with GPT-6 Astra. These project checks are not independent expert peer review. Ricardo Maldonado is the named author of this research release.

Version: TOE-N00AK-r1. Preparation date: 4 September 2026. This PDF is a release overview; the accompanying technical note supplies the full formulas and calculation protocol.